

SALMA EL BARDI

Data Science & Artificial Intelligence Engineering Student

Seeking an End-of-Studies Internship (PFE) – Starting January 2027

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PROFESSIONAL SUMMARY

Final-year Data Engineering & AI student at ENSA Fez. Hands-on expertise in designing scalable data pipelines (ETL, streaming, Lakehouse/Medallion architectures) and developing AI solutions (multi-agent systems, LLMs, Machine Learning). Seeking a 6-month end-of-studies internship (PFE) starting January 2027 in Data Engineering, Data Science, or AI.

EDUCATION

Engineering Degree in Data Science & AI — **2024 – Present**
National School of Applied Sciences (ENSA) — *Fez, Morocco*

Integrated Preparatory Cycle — **2022 – 2024**
National School of Applied Sciences (ENSA) — *Fez, Morocco*

High School Diploma (Baccalaureate) – Highest Honors — **2021 – 2022**
Independent Candidate: Physics & Chemistry Major — *Fez, Morocco*

TECHNICAL SKILLS

Data Engineering & Big Data: Apache Spark, PySpark, Kafka, Hadoop, Airflow, dbt, Medallion Architecture, ETL/ELT, Talend, Apache Hop, MinIO.

AI, LLMs & Data Science: Python, Scikit-learn, TensorFlow, PyTorch, Large Language Models (LLM), RAG, LangChain, CrewAI, ChromaDB, Multi-Agent Systems, NLP, Computer Vision, Recommender Systems.

Databases & BI: SQL, PostgreSQL, Microsoft SQL Server, Dimensional Modeling, Data Warehouse, Data Lakehouse, Power BI, Microsoft Fabric.

Cloud, MLOps & DevOps: AWS, Docker, Git, MLflow, FastAPI, Prometheus, Grafana, Streamlit.

WORK EXPERIENCE

Data Engineering & AI Intern – Acacia Afrique Système — June – August 2026

- **Data Processing & BI:** Extracted and transformed enterprise ERP data hosted on SQL Server; developed automated ETL workflows in Apache Hop with upstream pre-aggregations to optimize Power BI dashboard performance; scheduled nightly runs and built executive dashboards tracking sales, finance, purchasing, and fixed assets.
- **Multi-Agent AI System:** Engineered a Python-based autonomous agent framework using CrewAI, Ollama, and FastAPI to automate email triage, execute parameterized SQL queries, and generate report deliverables.

Full Stack Development Intern – ABH Sebou — July 2025

- Contributed to the development of a web application managing well-drilling permit authorizations for administrative staff and public users.

FEATURED PROJECTS

Tech Job Market Intelligence Lakehouse Pipeline — ***PySpark, Airflow, MinIO, Docker, Power BI***

- Engineered a containerized **Medallion Architecture** on **Docker**: multi-source ingestion, distributed batch processing with **PySpark**, **SHA-256** content-hash deduplication, and optimized **Snappy-compressed Parquet** storage on **MinIO S3**; delivered interactive hiring trend insights via **Power BI** dashboards.

Hybrid-TicketClassifier — ***MLflow, MiniLM, FastAPI, Prometheus, Ollama***

- Designed an asymmetric cascade routing architecture for banking support requests: absorbed most queries via a low-latency supervised ML Fast-Path and routed low-confidence queries to a local LLM Slow-Path (Llama); conducted an MLflow benchmark comparing TF-IDF and MiniLM dense representations, managed via Model Registry; deployed a FastAPI serving layer with real-time Prometheus latency tracking.

Data Warehouse & BI – Railway Industry (SNCF) — ***Talend, SQL Server, SSAS, Power BI***

- Built an enterprise **Data Warehouse** on **SQL Server** using a **star schema** (facts and dimensions); automated complex **ETL** flows using **Talend** and deployed an **OLAP cube** via **SSAS** (**MDX**); created an analytical **Power BI** dashboard (**DAX**) to assess route profitability and quantify cancellation impacts.

E-Commerce Real-Time Streaming Medallion Pipeline — ***Kafka, Docker, PostgreSQL, Power BI***

- Built an event-driven pipeline using **Apache Kafka** structuring stream data across **Bronze**, **Silver**, and **Gold** layers; stored clean data in **PostgreSQL** and applied **K-Means clustering** combined with **RFM customer segmentation** to detect at-risk profiles on a **Power BI** dashboard.

Churn Sentinel – Customer Attrition Prediction & Retention — ***Python, Scikit-learn, Streamlit***

- Built an end-to-end **Machine Learning** pipeline to forecast customer churn: preprocessing, feature encoding, and training a **Logistic Regression** model tuned for high sensitivity ; deployed an interactive **Streamlit** web interface for real-time risk scoring and recommended retention interventions.

CERTIFICATIONS

DataCamp: Docker and Kubernetes (2026) **365 Data Science:** Machine Learning Process A-Z (2025); Data Visualization (2024)
Oracle: Database Design & SQL (2024) **Coursera:** Python for Data Science, AI & Development (2024)

LANGUAGES & SOFT SKILLS

Languages: Arabic (Native), French (Fluent), English (Professional Working Proficiency).

Soft Skills: Analytical thinking, cross-functional collaboration, adaptability, technical communication, intellectual curiosity.